



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General handout for all courses appended to the timetable) this portion gives further details regarding the course.

Course No. : CHEM F333
Course Title : Chemistry of Materials
Instructor-in-charge : MADHUSHREE SARKAR

1. Course Description:

Solid state structure : unit cells, metallic crystal structures, polymorphism and allotropy, crystallographic direction and planes, closed packed crystal structures, polycrystalline materials, anisotropy; meso and micro porous materials: zeolites, composites, synthesis, characterization (XRD, SEM, TEM, AFM, FTIR, NMR, TGA, and DTA) and applications; ceramics and glass materials: crystalline and non-crystalline nature, glass-ceramics, processing; polymers: synthesis, structure, properties, inorganic polymers; mechanical properties: stress and strain, elastic and tensile properties, hardness, phase transformations, microstructure, alteration of mechanical properties; magnetic properties: atomic magnetism in solids, the exchange interaction, classification of magnetic materials, diamagnetism, pauli paramagnetism, ferromagnetism, antiferromagnetism, ferrimagnetism, superparamagnetism, ferromagnetic domains, hysteresis loop, hard and soft ferrites, applications; electrical properties: conductivity, band theory, types of semiconductors, time dependence of conductivity, mobility of charge carriers, metal-metal junction, metal-semiconductor junction, n-type and p-type semiconductors; optical properties: refraction, reflection, absorption, transmission, luminescence, photoconductivity, opacity and translucency in insulators, optical fibers; thermal properties: heat capacity, thermal expansion, conductivity, thermal stresses; corrosion: electrochemistry of corrosion of metals, different forms, environmental effects, prevention.

2. Scope and Objective of the Course:

- To provide a comprehensive understanding of crystalline structures, crystal systems, unit cells, crystallographic directions and planes, defects, and structure determination techniques.
- To develop knowledge of diffusion mechanisms, phase diagrams, phase equilibria, phase transformations, and their influence on microstructural evolution in materials.
- To establish the relationship between processing, microstructure, and mechanical properties of metals, including stress-strain behavior, elasticity, tensile properties, hardness, and deformation mechanisms.
- To introduce the structure and properties of ceramics, polymers, and other engineering materials, along with concepts of degradation, corrosion, electrical, and magnetic behavior.
- To enable students to analyze material structures, interpret phase and microstructural information, and correlate structure-property relationships for engineering material design and applications.

3. Text Books

T1. **Materials Science and Engineering - An Introduction** by William D. Callister, 10th Edition

4. Reference books:

R1. **Solid State Chemistry** by Lesley E. Smart and Elaine A. Moore

R2. **Materials Science and Engineering** by Donald R. Askeland and Pradeep P. Fulay

R3. **Materials Science for Engineers** by James F. Shackelford and Madanapalli K. Muralidhara



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5. Course Plan:

Module No.	Lecture Session	References
1. Structure of Crystalline Solids	L1.1: Unit Cells, Metallic Crystal Structures L1.2: Density Computations, Polymorphism and Allotropy, Crystal Systems L1.3: Point Coordinates L1.4: Crystallographic Directions L1.5: Crystallographic Planes	T1: 3.1-3.10
Learning outcome: - Identify and describe different crystal structures, unit cells, metallic crystal arrangements, and crystal systems. - Calculate basic crystal parameters, including density, lattice parameters, and atomic packing characteristics. - Represent atomic positions using point coordinates and determine crystallographic directions and planes. - Explain the concepts of polymorphism and allotropy and their influence on material structures and properties. - Analyze the relationship between crystal structure and the physical properties of crystalline materials.		
2. Ceramic Crystals and Crystal Structure Determination	L2.1: Linear and Planar Densities, Close- Packed Crystal L2.2: Ceramic Crystal Structures L2.3: Single Crystals, Polycrystalline Materials, Anisotropy, Non-crystalline Solids L2.4: Crystal Structure Determination by X-Ray Diffraction	T1: 3.11-3.17 T1: 12.2-12.7
- Explain ceramic crystal structures, close-packed arrangements, single and polycrystalline materials, anisotropy, and non-crystalline solids. - Calculate linear and planar densities and analyze the relationship between crystal structure and material properties. - Interpret X-ray diffraction (XRD) patterns to determine crystal structures		
3. Imperfections in Solids	L3.1: Vacancies and Self-Interstitials, Impurities in Solids L3.2: Specification of Composition L3.3: Dislocations– Linear Defects L3.4: Interfacial, Bulk or Volume Defects, Grain Size Determination	T1: 4.1-4.7, 4.11
- Identify and explain different types of crystal defects, including vacancies, self-interstitials, impurities, dislocations, and grain boundary defects. - Calculate atomic composition and describe the influence of defects on the structure and properties of solid materials. - Analyze the atomic arrangement near grain boundaries, twin boundaries, and other interfacial defects in crystalline materials.		
4. Diffusion in Materials	L4.1: Diffusion mechanisms L4.2: Steady-State Diffusion L4.3: Nonsteady-State Diffusion L4.4: Factors affecting Diffusion	T1: 5.1-5.5





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- Explain the fundamental mechanisms of diffusion and the factors influencing atomic movement in solid materials. - Differentiate between steady-state and non-steady-state diffusion processes and analyze their characteristics.		
5. Unary, Binary and Ternary Phase Diagrams	L5.1: Solubility Limit, Phases, Microstructure, Phase Equilibria, Gibbs Phase Rule, One-Component Phase Diagrams L5.2: Binary Isomorphous Systems, Interpretation of Phase Diagrams L5.3: Development of Microstructure in Isomorphous Alloys, Mechanical Properties of Isomorphous Alloys L5.4: Binary Eutectic Systems, Development of Microstructure in Eutectic Alloys, Equilibrium Diagrams Having Intermediate Phases or Compounds, Eutectoid and Peritectic Reactions, Congruent Phase Transformations L5.5: Ceramic and Ternary Phase Diagrams, Iron–Iron Carbide Phase Diagram, Development of Microstructure in Iron–Carbon Alloys	T1: 9.1-9.19
- Explain phase equilibria, solubility limits, phases, microstructures, and the application of Gibbs phase rule in material systems. - Interpret and construct unary, binary, and ternary phase diagrams, including isomorphous, eutectic, eutectoid, and peritectic systems. - Analyze phase transformations, determine phase compositions, and relate phase diagrams to microstructural development and mechanical properties of alloys.		
6. Phase Transformations, Microstructural Changes in Iron–Carbon alloys	L6.1: Kinetics of Phase Transformations, Metastable Versus Equilibrium States L6.2: Isothermal Transformation Diagrams L6.3: Continuous Cooling Transformation Diagrams	T1: 10.1-10.6
- Explain the kinetics of phase transformations and distinguish between equilibrium and metastable states in materials. - Interpret and construct isothermal transformation (TTT) and continuous cooling transformation (CCT) diagrams to analyze phase transformation behavior. - Relate transformation mechanisms to the development of microstructures and properties of iron–carbon alloys.		
7. Mechanical Properties of Metals	L7.1: Stress and Strain L7.2: Anelasticity, Elastic Properties of Materials L7.3: Tensile Properties L7.4: True Stress and Strain, Elastic Recovery After Plastic Deformation, Compressive, Shear, and Torsional Deformations, Hardness	T1: 6.1-6.10
- Explain fundamental mechanical properties of metals, including stress, strain, elasticity, plastic deformation, and the concepts of Hooke's law and Poisson's ratio. - Analyze tensile behavior, true stress–strain relationships, elastic recovery, hardness, and different modes of deformation such as compression, shear, and torsion. - Evaluate the relationship between mechanical behavior, material structure, and performance under different loading conditions.		
8. Polymer Structures	L9.1: Chemistry of Polymer Molecules, Molecular Weight, Molecular Shape, Molecular Structure, Molecular	T1: 14.3-14.11





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	Configurations L9.2: Thermoplastic and Thermosetting Polymers, Copolymers, Polymer Crystallinity	
	- Explain the chemistry, molecular structure, molecular weight, configurations, and shapes of polymer molecules. - Differentiate between thermoplastic, thermosetting polymers, copolymers, and their structural characteristics. - Relate polymer crystallinity and molecular structure to the physical properties and applications of polymeric materials.	
9. Electrical Properties	L11.1: Energy Band Structures in Solids, Conduction in Terms of Band and Atomic Bonding Models, Electron Mobility, Electrical Resistivity of Metals, L11.2 Intrinsic and Extrinsic Semiconduction, Temperature dependence of Carrier Concentration L11.3 Factors Affecting Carrier Mobility, Hall Effect	T1: 18.5-18.8, 18.10-18.14
	- Explain the electrical behavior of solids based on energy band structures, atomic bonding models, electron mobility, and electrical resistivity. - Differentiate between metallic conduction, intrinsic and extrinsic semiconductors, and analyze the factors affecting charge carrier concentration and mobility. - Apply concepts such as temperature dependence of conductivity and Hall effect to understand the electrical properties of metals and semiconductor materials.	
10. Magnetic Properties	L12.1: Diamagnetism and Paramagnetism, Ferromagnetism Antiferromagnetism and Ferrimagnetism L12.2: Influence of Temperature on Magnetic Behavior Domains and Hysteresis Magnetic Anisotropy	T1: 20.1-20.8
	- Explain the origin of magnetism in solids and classify materials based on their magnetic behavior, including diamagnetic, paramagnetic, ferromagnetic, antiferromagnetic, and ferrimagnetic materials. - Describe the influence of temperature, magnetic domains, hysteresis behavior, and magnetic anisotropy on the properties of magnetic materials. - Relate magnetic structure and properties to the performance and applications of magnetic materials.	





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6. Evaluation Scheme:

Component	Duration	Weightage (%) Marks (100)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30 (30)	As per AUGSD	Closed Book
Comprehensive Examination	180 Min.	40 (40)	As per AUGSD	Closed and Open Book
Continuous Evaluations	15 min.	30 (30)	Continuous	Closed Book

For continuous evaluation, a total of four quizzes will be conducted, of which the best three scores will be considered for the final evaluation.

For students with ID numbers 2024B2XXXX and 2025B2XXXX, one of the four quizzes will be designated as an attendance-linked quiz. This quiz will be graded only for students who maintain a minimum attendance of 50%. For these students, the final quiz component will be based on the attendance-linked quiz (if eligible) and the best two scores from the remaining three non-attendance-linked quizzes.

7. Minimum criterion for awarding valid grades: Where a student continued to remain registered in a course but gave the instructor inadequate opportunity to evaluate him/her by absenting himself/herself from quizzes/ tests/ examinations/other components of evaluation, or by appearing in the same for the sake of appearance without applying himself/herself to the task in hand or by submitting a blank script (answer book) this event will be reported as NC (Not Cleared).

A student should obtain 30% of the average of the top 10 performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC. This will be applicable to students of ID No. 2024B2XXXX and 2025B2XXXX.

8. Consultation Hour:

9. Notices: Notices, if any, will be displayed on **Nalanda**.

10. Make-up Policy: Make-up would be considered only for **genuine cases**.

11. Note (if any): -

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